Kepler's Third Law and Body–Flow Coupling

A Cavity Hypothesis and Conditional Compatibility Tests

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Abstract

The author hypothesizes that a common coupling between material bodies and a proposed inward-moving non-material medium underlies weight and inertial response. A velocity-field interpretation must explain both sustained body–field comotion and the acceleration of a body initially released with a different velocity. We test these conditions together for an instantaneous linear coupling $\dot{\mathbf{v}} = D_u\mathbf{u} - K(\mathbf{v} - \mathbf{u})$, where K is symmetric positive semidefinite. The exact slip equation includes the velocity gradient and therefore distinguishes passive damping from actual slip contraction. Applying a standard positive-operator compatibility argument, we find that for a specified nonzero release slip $\mathbf{w}_0$ and acceleration discrepancy $\mathbf{d}$, a passive operator satisfying the release requirement exists if and only if $\mathbf{w}_0 \cdot \mathbf{d} > 0$, or $\mathbf{d} = 0$. A rank-one construction and a lower bound on the unavoidable residual follow. Applying the criterion to a specified incompressible, rotating poloidal inflow gives both locally admissible points and an analytic obstruction at a fixed-normal outer equatorial boundary. Because a passive operator damps uniform relative motion, a second, non-dissipative comparison class is examined, in which the body responds only to changes of motion. It gives free fall $\dot{\mathbf{v}} = \kappa\,\mathbf{a}_u$ with one coefficient shared by weight and inertial response, independent of cavity size. Comotion and free fall hold together only for $\kappa = 1$, which requires the inflow profile $u_r^2 r = 2E$. For $\kappa = 2$ the required inflow speed equals the circular orbital speed at every radius. The cumulative front matches the inverse-square law in this class only at one radius. The author's working interpretation treats material atoms as cavities carried through the proposed medium. It does not adopt a rigid-body inertia law as its mechanism. The classical spin-budget calculation is retained only as an excluded comparison in an appendix. The passive-response results likewise describe a restricted diagnostic class, not a derived material-conversion law. They do not identify the proposed mechanism in nature. Fifty recorded computational checks verify the stated algebraic and local numerical claims. The article provides testable compatibility conditions rather than a claimed experimental detection of a non-material medium.

Keywords: material derivative, velocity slip, passive coupling, positive semidefinite operator, angular momentum, free fall.

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1 Introduction: two requirements for one process

The proposed non-material medium, to which the author attributes properties of an ideal incompressible fluid or gas with zero internal viscosity, is called the E-field in the companion study [4].

Main results. The study applies a constructive positive-operator criterion to the compatibility of comotion and released-body acceleration under one prescribed response law. It derives a separate strain condition for decay of velocity differences and evaluates both criteria in a rotating three-dimensional inflow. A non-dissipative comparison class expresses response to flow acceleration and to changes of relative motion through one coefficient, providing a precise condition for universal response in that class.

The orbital programme [1, 2] and the radial interpretation [3] motivate a joint question. Can one medium account for a body moving with its local flow on an established orbit and for a body released from rest relative to the rotating surface? These initial conditions are different. The acceleration must be evaluated for each through the same constitutive law. Assuming $\mathbf{v} = \mathbf{u}$ in one case and imposing $\ddot{r} = -E/r^2$ independently in another does not derive their connection.

The parameters of selected celestial sources and controlled combination of their fields are examined in the companion multiple-system study [5]. Its body-motion calculation and the medium diagnostics here address different parts of the same question. Connecting them requires an explicit coupling law, and agreement of orbital equations alone does not establish the medium mechanism.

The present study applies a local positive-operator compatibility criterion to body–flow response. The companion transition study constructs and tests the candidate flow used here [6]. The atom-as-cavity interpretation motivates the inquiry, without supplying an experimentally established constitutive law. Ordinary fluid density and stress in the classical appendix are conditional continuum quantities, not assigned measurements of the vacuum. The tensor balance and torque-free control in that appendix are comparison calculations, rather than the mechanism of the proposed medium.

Separation of the proposal and the diagnostic model. The working hypothesis distinguishes an internally frictionless non-material medium from its coupling to material bodies, interpreted here as assemblies of atomic cavities. The author interprets the difference between the cumulative flow kinematics and the body's falling motion as an indication of that coupling, proposed to underlie both weight and inertial response. Interactions between atoms are a separate question. The quantitative coupling law remains to be derived. Here "friction" refers to the interaction between the flow and atoms rather than a specified viscous drag law. The orbital relation $E = \omega^2 R^3$, together with the cumulative radial relation $v^2 R^3 = C$ for a linear inward speed v [3, 6], motivates the proposal. These relations do not derive the quantitative coupling law or universal body response. The passive operator below is an explicitly auxiliary diagnostic. The author's second-phase target is $\mathbf{w} = \mathbf{v} - \mathbf{u} = 0$ for the test body and its local medium, with their common inward motion described in the central body's corotating frame. The medium flows around impermeable atomic cavities, including through the open centre of toroidal forms. It does not enter the cavities. Formation of new atomic standing waves is a separate question, not a passage or conversion inside existing atoms. This velocity target neither specifies a viscous drag law nor removes the proposed body–medium interaction.

The companion study of external flow around atoms [14] examines pressure-gradient response with effective standing-wave inertia and a possible additional relative-velocity term. The impermeable atomic boundary excludes the net-inflow premise of a point-sink model. The pressure

response and the added response to body acceleration provide the principal comparison relevant here.

2 Related work and modelling domain

Particle motion in nonuniform fluids has an established theory. Maxey and Riley [9] treat a small rigid sphere, including effects not represented by a single drag coefficient. Talaei and Garrett [11] discuss extensions and limitations beyond the simplest regime. These works concern material fluids and specified particles. They cannot be transferred to a non-material medium without establishing which, if any, of their constitutive assumptions are applicable.

Our instantaneous operator is a restricted diagnostic model. Its advantages are that existence and impossibility can be stated exactly, and that it separates release acceleration from contraction of velocity differences. It omits memory, finite-size response, rotational degrees of freedom of the test particle, nonlocal stresses and nonlinear constitutive effects. A negative result below applies to this domain and the stated flow boundaries, not to every imaginable flow interpretation.

3 The working cavity/flow proposal and directional tests

In the author's hypothesis, atoms are collapsed three-dimensional standing waves, bubble-like voids which the non-material medium cannot enter. An aggregate of atoms, a material body, creates a large pressure difference for this medium. Under its own pressure the medium flows from regions of higher pressure to regions of lower pressure, through different stages of vortex and coupled motion. It flows around atoms on all sides, including through the open centre of a torus-shaped atom. Passage through that opening is not penetration into the inaccessible atomic cavity. The medium has the properties of an ideal incompressible fluid or gas with zero internal viscosity. Where the available flow channels narrow, continuity requires an increase of local speed. The proposed interaction concerns this accelerated external flow around atoms. The formation of standing waves and the detailed conversion mechanism are outside the present study. The author interprets material atoms as impermeable cavities transported through a non-material medium. Apparent inertial behaviour is proposed to arise from that transport rather than an intrinsic rigid-body inertia law. This is the working hypothesis, not an established measurement of vacuum density. The author tentatively suggests little dependence on speed magnitude but a possible dependence on direction. Neither dependence has yet been quantified. The reference direction must therefore be stated before testing a prediction: local flow and the central rotation axis are distinct possible reference vectors. No speed-independent response or anisotropy coefficient is fitted here.

The comoving target is $\mathbf{w} = \mathbf{v} - \mathbf{u} = 0$. With sufficiently smooth fields, this implies the kinematic identity $\dot{\mathbf{v}} = \partial_t \mathbf{u} + (\mathbf{u} \cdot \nabla)\mathbf{u}$ along that path. It contains no inertia parameter, but does not determine how comotion is established or how a cavity is converted or transported. The passive operator examined below is explicitly a comparison class, not an identification of the author's interaction with passive drag.

4 Velocity, material derivative and slip evolution

Let $J = \nabla \mathbf{u}$ with $J_{ij} = \partial_j u_i$, and $\mathbf{a}_u = \partial_t \mathbf{u} + J\mathbf{u}$. Consider

$$\dot{\mathbf{x}} = \mathbf{v}, \qquad \dot{\mathbf{v}} = \mathbf{a}_u - K\mathbf{w}, \qquad \mathbf{w} = \mathbf{v} - \mathbf{u}(\mathbf{x}, t), \quad K = K^T \succeq 0. \tag{1}$$

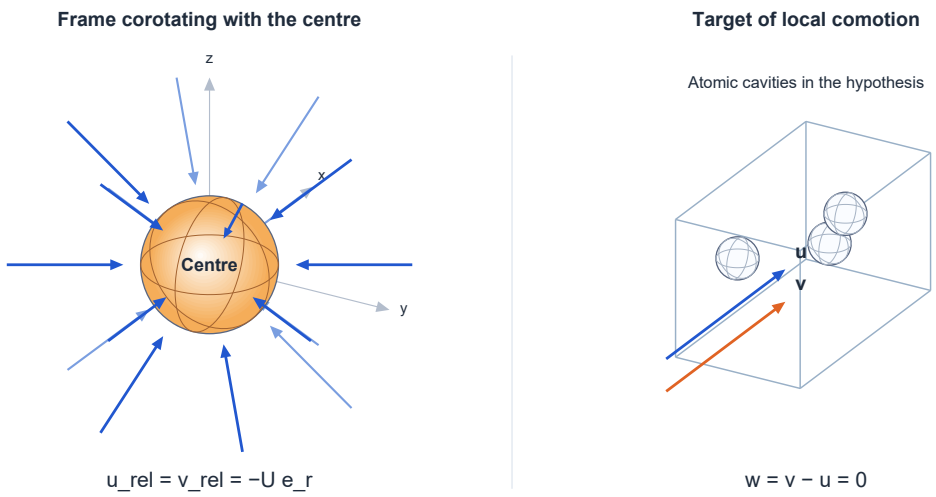


Figure 1: Author-proposed distinction between the central corotating description and local body–medium comotion. Arrows specify the target velocity relation only. Cavity geometry, density, speed independence and directional response are not observationally established by this schematic.

Along the body, $\ddot{\mathbf{x}}$ and $D_u\mathbf{u}$ use different advective velocities. Subtracting the total derivative of $\mathbf{u}(\mathbf{x}, t)$ gives

$$\dot{\mathbf{w}} = -(J + K)\mathbf{w}. \tag{2}$$

Thus zero slip is invariant for a sufficiently regular prescribed flow, while nonzero slip is affected by strain as well as damping. Zero slip does not imply a circular orbit when the field itself has inward velocity.

At release, take $\mathbf{v}_0 = \boldsymbol{\Omega}_b \times \mathbf{r}$ relative to the centre. The kinematic target is the operational central acceleration $\mathbf{g} = -E\mathbf{r}/r^3$, not yet a derived consequence of Eq. (1). Define

$$\mathbf{w}_0 = \mathbf{v}_0 - \mathbf{u}, \qquad \mathbf{d} = \mathbf{a}_u - \mathbf{g}. \tag{3}$$

Reproducing that target is exactly $K\mathbf{w}_0 = \mathbf{d}$. Material independence would additionally require that the same prescribed response and initial state give the same acceleration for different body compositions. Constructing a separate fitted K for each body would not meet that requirement.

5 Applying a positive-operator criterion to passive release

The strict condition for a positive-definite symmetric matrix mapping a nonzero vector s to a vector y is the familiar curvature condition $s^T y > 0$ in secant-matrix constructions [12, 13]. Nocedal and Wright state the necessary condition and existence of such matrices in Section 6.1. Here it is applied to the release slip and acceleration discrepancy, with positive semidefiniteness allowing the additional zero-discrepancy case. The elementary proof below includes that case and characterizes all passive solutions. The criterion itself is not presented as a new physical law or a new optimization result. Its use here concerns local body–flow compatibility and the boundary obstruction derived below.

For $\mathbf{w}_0 \neq 0$, a symmetric positive-semidefinite K with $K\mathbf{w}_0 = \mathbf{d}$ exists if and only if

$$\sigma = \mathbf{w}_0 \cdot \mathbf{d} > 0 \quad \text{or} \quad \mathbf{d} = 0. \tag{4}$$

Necessity follows from $\sigma = \mathbf{w}_0^T K \mathbf{w}_0 \geq 0$. For a positive-semidefinite matrix, a zero quadratic form implies $K\mathbf{w}_0 = 0$, so the borderline case also requires $\mathbf{d} = 0$. For $\sigma > 0$ a constructive solution is

$$K_0 = \frac{\mathbf{d}\mathbf{d}^T}{\sigma}. \tag{5}$$

All positive-semidefinite solutions are $K = K_0 + C$, where $C \succeq 0$ and $C\mathbf{w}_0 = 0$. The Schur-complement argument is given in Appendix A. When $\mathbf{w}_0 = 0$, the target can be satisfied only if $\mathbf{d} = 0$.

Equation (5) is an algebraic witness, not a derived constitutive law. It uses the desired acceleration to construct the response and is therefore not an independent prediction of that acceleration. For any passive K, a useful obstruction bound is

$$|\mathbf{d} - K\mathbf{w}_0| \geq \max\{0, -\sigma/|\mathbf{w}_0|\}. \tag{6}$$

It follows by projecting the residual onto the slip direction. The norm is an acceleration, so a comparison with E/r^2 has an explicit physical normalization.

6 Passivity does not guarantee convergence to the flow

Let $S = (J + J^T)/2$. The local slip-energy rate is

$$\frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}t}|\mathbf{w}|^2 = -\mathbf{w}^T(S + K)\mathbf{w}. \tag{7}$$

Consequently $K \succeq 0$ alone need not contract every slip direction. On the positive branch in Eq. (4), the release direction fixes

$$\eta = \frac{\sigma + \mathbf{w}_0^T S \mathbf{w}_0}{|\mathbf{w}_0|^2}. \tag{8}$$

Strict instantaneous contraction in every direction is possible with a compatible K if and only if $\eta > 0$. Necessity follows along $\mathbf{w}_0$. Sufficiency follows by adding a large positive multiple of the projector orthogonal to $\mathbf{w}_0$. The zero case requires a separate semidefinite analysis and is not called strict contraction.

These are instantaneous statements. If along a future trajectory $S + K \succeq \lambda(t)I$, then

$$|\mathbf{w}(t)| \leq |\mathbf{w}(0)| \exp\left[-\int_0^t \lambda(s)\,\mathrm{d}s\right]. \tag{9}$$

A positive constant lower bound would give a tolerance-reaching time $t \geq \lambda^{-1}\ln(|\mathbf{w}(0)|/\varepsilon)$. No astronomical value of λ has been inferred here. Smooth linear evolution also does not generally make nonzero slip exactly zero in finite time.

7 Application to a rotating three-dimensional inflow

7.1 Test field and initial conditions

Use the divergence-free shell field developed in the companion transition study [6], $a \leq r \leq S_o$,

$$u_r = -A/r^2 - Br^{-3/2}P_2(\cos\theta), \tag{10}$$

$$u_\theta = (B/4)r^{-3/2}\sin\theta\cos\theta, \tag{11}$$

$$u_\phi = r\Omega(r)\sin\theta. \tag{12}$$

The symbol S_o denotes the outer radius, whereas S denotes strain above. For $q = B\sqrt{r}/A$, all-direction inflow requires $-1 < q < 2$. The pressure calculation used to define the initial $\partial_t \mathbf{u}$ satisfies incompressible Euler acceleration and fixes $\partial_t u_r = 0$ at both spherical boundaries. It is an initial compatibility calculation, not a computed long-time evolution. The pressure prescription and smooth rotation profile are part of the shared software material described in Data and code availability.

7.2 A boundary obstruction independent of the interior interpolator

At the outer equator, require $\Omega(S_o)^2 S_o = E/S_o^2$ and use an instantaneous corotating release with $w_{0\phi} = 0$. This diagnostic release isolates the radial condition. It does not assert corotation of the actual body at every outer radius. There $u_\theta = 0$ and

$$U = u_r = -\frac{A}{S_o^2} + \frac{B}{2S_o^{3/2}} < 0, \quad D = \partial_r u_r = \frac{A}{S_o^3}\left(2 - \frac{3}{4}q\right) > 0. \tag{13}$$

The centrifugal part of the material acceleration equals the prescribed central target. The remaining radial discrepancy is $d_r = UD$ and

$$\sigma = -U^2 D < 0. \tag{14}$$

Axisymmetric pressure has no azimuthal gradient, so $d_\phi = 0$. Even if a surface-corotating release has nonzero azimuthal slip there, it does not alter this dot product. Hence this fixed-normal, all-inward boundary class cannot satisfy the passive release condition throughout the shell. Allowing time-dependent normal velocity changes the necessary radial condition to $\partial_t u_r \geq -UD > 0$, with the zero-dot-product case still requiring the full vector discrepancy to vanish. This identifies a specific boundary assumption to change rather than rejecting all coupling mechanisms.

7.3 A reproducible nondimensional example

Take $a = A = E = \Omega_b = 1$, $S_o = 4$, $B = 0.9$. Then $q(S_o) = 1.8$, $U = -0.00625$, $D = 0.01015625$, and $\sigma = -3.9672852 \times 10^{-7}$ at the outer equator. At the inner pole, $w_{0r} = 1.9$, $(a_u)_r = -6.365$, $g_r = -1$, so $\sigma = -10.1935$. Equation (6) yields an unavoidable acceleration residual of at least 5.365 in these units, larger than the target magnitude one.

Table 1: Local release and contraction diagnostics for the example.

r	θ (degrees)	σ	η	Passive release / strict contraction
1	0	-10.1935	0.526316	no / not applicable
1.2	17.1887	-2.022542	0.636331	no / not applicable
2	45	0.00495305	-0.209550	yes / no
4	0	0.00868848	0.357143	yes / yes locally

At the outer pole, one compatible operator is $K = \mathrm{diag}(0.0734375, 0.0734375, 0.28370536)$ and $S + K = \mathrm{diag}(0.03671875, 0.03671875, 0.35714286)$ in Cartesian axes with the pole on the third axis. A locally successful construction is not a global constitutive model. Fifty recorded checks passed, with an independent Jacobian comparison below 1.85×10^{-10}.

8 A non-dissipative coupling class: response to changes of motion

8.1 Classical motivation and comparison assumptions

The passive operator of Eq. (1) acts on the velocity difference $\mathbf{w}$. Such a term damps uniform relative motion, which the author's proposal excludes: the medium has no internal friction and

does not brake uniform motion. A second comparison class has exactly that property. It is motivated by the acceleration terms in the small-sphere equation of Maxey and Riley [9] and by the inviscid potential-flow result for a sphere of fixed volume [10], with no drag, volume growth or external volumetric term. The classical reference assumes an incompressible medium and negligible interactions between neighbouring particles. Here the acceleration law is used as a restricted comparison, not as a derivation for the proposed non-material medium, which is assigned ideal incompressible fluid or gas properties.

8.2 The law

Let a body of volume V move in a medium of conditional density ρ. Three reactions are retained, each proportional to ρV: the reaction of the accelerating medium, $\rho V \mathbf{a}_u$, the reaction of the medium set in motion around the body, $c_a \rho V(\mathbf{a}_u - \dot{\mathbf{v}})$, with a shape coefficient $c_a > 0$ ($c_a = 1/2$ for a sphere). A constant scalar c_a is assumed here. General nonspherical bodies require an added-inertia tensor rather than this scalar law. The third contribution is the body's own inertia, $\beta \rho V \dot{\mathbf{v}}$, where $\beta \geq 0$ compares the body's own inertia with that of the displaced medium. For an empty classical cavity with negligible intrinsic response $\beta = 0$. This is not an assigned value for an atomic standing wave. The author proposes the effective condition $\beta = 1$ for unit response, whose microscopic derivation remains open. Balancing,

$$\beta \dot{\mathbf{v}} = \mathbf{a}_u + c_a(\mathbf{a}_u - \dot{\mathbf{v}}), \qquad \dot{\mathbf{v}} = \kappa\, \mathbf{a}_u, \qquad \kappa = \frac{1 + c_a}{\beta + c_a}. \tag{15}$$

The volume V and the conditional density ρ cancel. For a sphere, $\kappa = 3$ for a pure cavity, $\kappa = 2$ for $\beta = 1/4$, and $\kappa = 1$ for $\beta = 1$.

8.3 Properties

No braking of uniform motion. Equation (15) contains no term in $\mathbf{w}$. Subtracting the total derivative of $\mathbf{u}(\mathbf{x}, t)$,

$$\dot{\mathbf{w}} = (\kappa - 1)\, \mathbf{a}_u - J\mathbf{w}. \tag{16}$$

Uniform relative motion through a uniform medium is therefore neither damped nor excited. A response appears only where the medium accelerates or the body's motion relative to the flow is changed.

One coupling for weight and inertial response. If the body is held at rest relative to the centre, the medium supplies $\rho V(1 + c_a)\, \mathbf{a}_u$, and the support must supply its opposite to maintain equilibrium. If the body is accelerated through a relatively stationary medium, the reaction to that change is $\rho V(\beta + c_a)\, \dot{\mathbf{v}}$. Both coefficients come from the same coupling, and the free-fall acceleration is their ratio, Eq. (15). Within this class, free fall is universal if and only if κ is the same for all bodies. For isolated geometrically similar cavities with $\beta \to 0$ and a common shape coefficient, this ratio does not depend on the cavity size. Extending that result to an assembly requires additive response, a common local acceleration and negligible mutual flow disturbances. Under those additional assumptions the number of identical cavities cancels from the ratio. Independence from their arrangement is not established by the single-cavity calculation. The measured universality of free fall then constrains differences of κ between materials, not κ itself [4].

Comotion and release together. The author's second-phase target is $\mathbf{w} = 0$. By Eq. (16), zero slip persists in an accelerating medium only if $\kappa = 1$. The release target is $\dot{\mathbf{v}} = \mathbf{g}$, which requires $\mathbf{a}_u = \mathbf{g}/\kappa$. Invariance of zero slip does not establish approach to the flow from a different initial velocity. For $\kappa = 1$, $\dot{\mathbf{w}} = -J\mathbf{w}$, so strain can amplify a nonzero difference. General convergence in arbitrary directions is not proved by this law. In the specified radial profile,

however, a body released at rest from r_0 at $\kappa = 1$ has $|v_r/u_r| = \sqrt{1 - r/r_0}$. It approaches the medium speed as r/r_0 decreases without velocity damping [14]. This is radial approach in a prescribed field, not exact comotion reached in finite time or a guarantee before impact with a finite source. For a steady radial inflow $u_r(r)$, $u_r\,\mathrm{d}u_r/\mathrm{d}r = -E/(\kappa r^2)$ gives

$$u_r^2\, r = \frac{2E}{\kappa} \tag{17}$$

when the inflow speed vanishes far away. With $\kappa = 1$, both targets hold together, and the inflow speed equals $\sqrt{2E/r}$, the profile of the river model discussed in the companion transition study [6]. With $\kappa = 2$, Eq. (17) becomes $u_r^2 r = E$: the radial inflow speed equals the circular orbital speed $\sqrt{E/r}$ at every radius, the orbital relation $E = U^2 R$ in radial form. With $\kappa = 3$ the inflow speed is $\sqrt{2E/3r}$. At the Earth's surface these three speeds are 11.186, 7.910 and 6.458 $\mathrm{km\,s^{-1}}$. A steady profile $u_r \propto r^{-1/2}$ is not divergence-free at constant density in a spherical domain [6]. For a coarse-grained medium of constant density, that profile requires a distributed volume-removal term in continuity. In the author's proposal such a term would describe formation of new atoms by collapse of the medium, not entry into existing atoms. Its quantitative law and the formation mechanism are outside this study. The divergence-free classical exterior and the conversion-averaged profile are distinct descriptions.

The cumulative front. For the cumulative law $v^2 R^3 = C_r$ with $C_r = E\ell^2$, the front acceleration is $-3E\ell^2/(2R^4)$ [6, 4]. If the local medium acceleration is identified with that front acceleration, this class assigns the body κ times that value. This is an additional identification, not a general equivalence of boundary and spatial velocities. The assigned acceleration equals $-E/R^2$ only at

$$R_\kappa = \ell\sqrt{3\kappa/2}, \tag{18}$$

that is at $1.224745\,\ell$, $1.732051\,\ell$ and $2.121320\,\ell$ for $\kappa = 1, 2, 3$. The local cumulative front cannot reproduce the inverse-square law at all radii through this coupling. The discrepancy between the two laws, which the author interprets as the signature of the coupling, is therefore not removed by an added-medium response alone.

8.4 Scope of the non-dissipative comparison class

The non-dissipative comparison class defined by Eq. (15) satisfies the requirements stated by the author that the passive class violates: no braking of uniform motion, a response only to changes of motion, and a single coupling behind weight and inertial response. It is a comparison class taken from ideal-fluid dynamics, not a derived law for a non-material medium. Its coefficient κ has not been determined for atomic cavities, and its required inflow profile, Eq. (17), is not the cumulative law. Determining whether a conversion law can reconcile the two is part of the remaining programme.

9 Conclusions

An exact compatibility criterion. We establish that, for one linear passive response, for a nonzero release slip $\mathbf{w}_0$, comotion and the acceleration of a released body are compatible if and only if $\mathbf{w}_0 \cdot \mathbf{d} > 0$, or $\mathbf{d} = 0$. At $\mathbf{w}_0 = 0$, compatibility requires $\mathbf{d} = 0$. A rank-one construction, a residual bound and a strain condition for contraction of velocity differences follow.

Applied to a rotating three-dimensional inflow. The criterion gives locally admissible points and an analytic obstruction at the fixed-normal outer equatorial boundary, $\sigma = -U^2 D < 0$. It thereby identifies the specific boundary assumption to change: a time-dependent normal velocity must satisfy the necessary condition $\partial_t u_r \geq -UD$. Equality is admissible only if the full

vector discrepancy vanishes. This boundary condition alone does not establish a global response law.

A non-dissipative class meeting the author's requirements. A response to flow acceleration and to changes of motion, without braking of uniform motion, gives $\dot{\mathbf{v}} = \kappa \mathbf{a}_u$ with one coefficient shared by weight and inertial response, independent of volume at fixed β and scalar c_a in the local potential-flow comparison. For a steady inward radial field vanishing at infinity, comotion and free fall hold together exactly for $\kappa = 1$, with inflow at the second characteristic speed, $u_r^2 r = 2E$. For $\kappa = 2$ the inflow equals the circular orbital speed. At $\kappa = 1$, a body released from rest approaches the medium speed as $\sqrt{1 - r/r_0}$.

Next step. The coarse-grained conversion balance for that profile, the author's collapse of the medium into new atoms, remains to be quantified, together with β for the atomic standing wave. The cumulative front matches the inverse-square law in this class only at $R_\kappa = \ell\sqrt{3\kappa/2}$, which locates where the two laws must be joined.

Declarations

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Competing interests. The author is affiliated with RWA® RAILWAYAUTOMATIC®, which supports the research programme. The author declares no other competing interests.

AI-based tools were used to assist with calculations, code development and manuscript preparation. The author takes full responsibility for the final content. No new experiment or observation is reported here.

Data and code availability

An earlier Ew2R3 visualization release is archived on Zenodo [7]. Calculations, input data and interactive visualization for this study are part of the Ew2R3 software release [8]. The programme website is Ew2R3.org.

The results are mathematical and computational. The shared release includes the dimensionless test field, pressure solution, derivative checks, operator witnesses and synthetic controls. No new observational sample is claimed. The original observations discussed as targets remain attributable to their cited providers.

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A All passive solutions and strict contraction

Choose an orthonormal basis with first vector $e = \mathbf{w}_0/|\mathbf{w}_0|$. The prescribed first column of K is $\mathbf{d}/|\mathbf{w}_0|$. For $\sigma > 0$, the first diagonal entry is $k = \sigma/|\mathbf{w}_0|^2 > 0$. Positivity of the block matrix is equivalent to positivity of its transverse Schur complement. The rank-one K_0 has zero Schur complement. Every other solution therefore adds precisely a transverse positive-semidefinite block, proving the form $K_0 + C$ with $C\mathbf{w}_0 = 0$.

For $S + K$, the first diagonal entry is η. If it is positive, adding $\gamma(I - ee^T)$ with sufficiently large γ makes the transverse Schur complement positive definite. If it is negative, no such addition changes the negative quadratic form along e. These proofs are pointwise and require no assumption of a measured medium. Integrating the energy inequality yields Eq. (9) only when the needed bound holds along the evolving path.

B Scope of the computational witness

The computational material covers the release criterion, pressure and boundary conditions, derivative comparison, matrix positivity, release residual and counterexamples. These tests concern different parts of the construction. They do not estimate a physical density, relaxation time or moment of inertia for the proposed non-material substance. The operator test must be rerun if the pressure or normal-boundary evolution changes. Its results cannot be carried to another field with changed conditions.

The separate tensor-deposition and three-axis rigid-body controls below test conditional classical identities. They do not increase the number of observations or identify the body–medium coupling.

C Conditional classical spin-budget comparison

C.1 Transport, deposition and the angular-momentum budget

This appendix records a calculation under ordinary continuum and rigid-body assumptions. Those assumptions are not transferred to the author's cavity/flow proposal. No conclusion here is imposed as its constitutive law. Let I and Ω_b describe an axisymmetric material body. An angular budget must specify what crosses the boundary and what stress transfers rotation:

$$\frac{\mathrm{d}}{\mathrm{d}t}(I\Omega_b) = \mathcal{J}_{\mathrm{in}} - \mathcal{J}_{\mathrm{out}} + \mathcal{T}_{\mathrm{stress}}. \tag{19}$$

This equation expresses transport and exchange. A velocity boundary condition alone does not determine any of its terms. In an ordinary continuum illustration, inflow corotating on a sphere of radius a brings specific angular momentum $\Omega_b a^2 \sin^2\theta$. If it is deposited at the same cylindrical lever arm, its contribution to I has exactly that factor without Ω_b. For the field above and constant density ρ,

$$\mathcal{J}_{\mathrm{in}} = \frac{8\pi\rho}{3}\Omega_b a^2\left(A - \frac{B\sqrt{a}}{5}\right) = \Omega_b \dot{I}_{\mathrm{in}}. \tag{20}$$

With only such deposition and no other transport or stress, $I\dot{\Omega}_b = 0$. Inward motion and angular-momentum input can therefore coexist without spin-up. This is a deposition bookkeeping identity under the stated assumptions, rather than a primary result about the proposed medium. Transport to a different lever arm, stress, removal of material or an independently changing internal I would change this result and must be modelled rather than implied by the words "inflow spins the body".

The companion transition study derives the following kinematic no-slip boundary condition in an axisymmetric Euler field [6]. It constrains the radial derivative of $\Omega(r)$. Writing $\alpha = a\Omega'(a)/\Omega_b$, latitude-independent boundary spin acceleration for $B\Omega_b \neq 0$ requires $\alpha = -5/3$ in this family, giving $\dot{\Omega}_b = A\Omega_b(2-q(a))/(6a^3)$. The discrepancy with the deposition-only budget is informative: enforcing a velocity evolution at the boundary does not provide the missing physical angular-momentum exchange.

C.2 Three-axis balance and the rigid-body limit

In a body-attached orthonormal frame, let $\mathsf{I}(t)$ be the inertia tensor and let $\boldsymbol{\Omega}$ have three components. The vector budget is

$$\mathsf{I}\dot{\boldsymbol{\Omega}} + \dot{\mathsf{I}}\boldsymbol{\Omega} + \boldsymbol{\Omega} \times (\mathsf{I}\boldsymbol{\Omega}) = \boldsymbol{\mathcal{J}}_{\text{in}} - \boldsymbol{\mathcal{J}}_{\text{out}} + \boldsymbol{\mathcal{T}}_{\text{stress}}. \tag{21}$$

The cross product converts the derivative of the components in the rotating frame to the inertial derivative of angular momentum. It is not an extra exchange mechanism. For changing shape the complete tensor derivative must be retained. Using a single scalar I would discard relevant information.

The deposition result also has a tensor form. In the same conditional continuum illustration, let $\mathrm{d}\nu \geq 0$ denote an inertial influx measure at a boundary position $\mathbf{r}$, deposited at that same position. Then

$$\dot{\mathsf{I}}_{\text{in}} = \int (r^2 1 - \mathbf{r}\mathbf{r}^T)\,\mathrm{d}\nu, \qquad \boldsymbol{\mathcal{J}}_{\text{in}} = \int \mathbf{r} \times (\boldsymbol{\Omega} \times \mathbf{r})\,\mathrm{d}\nu = \dot{\mathsf{I}}_{\text{in}}\boldsymbol{\Omega}. \tag{22}$$

This follows from the vector triple-product identity and does not require an axisymmetric influx. If this is the only change of I, with no outflow or stress transport, the two deposition terms cancel in Eq. (21). What remains is $\mathsf{I}\dot{\boldsymbol{\Omega}} + \boldsymbol{\Omega} \times (\mathsf{I}\boldsymbol{\Omega}) = 0$. For a nonspherical body this does not imply constant components of $\boldsymbol{\Omega}$. The scalar no-spin-up statement above applies to its axisymmetric single-axis limit. The influx measure has not been inferred for a non-material medium. Energy and boundary evolution would still need closure.

For a rigid body with constant principal moments and zero exchange, the same balance gives the standard three equations, not a new medium-specific law:

$$I_1\dot{\Omega}_1 + (I_3 - I_2)\Omega_2\Omega_3 = 0, \tag{23}$$

$$I_2\dot{\Omega}_2 + (I_1 - I_3)\Omega_3\Omega_1 = 0, \tag{24}$$

$$I_3\dot{\Omega}_3 + (I_2 - I_1)\Omega_1\Omega_2 = 0. \tag{25}$$

They conserve $\frac{1}{2}\boldsymbol{\Omega}^T\mathsf{I}\boldsymbol{\Omega}$ and $|\mathsf{I}\boldsymbol{\Omega}|^2$. In a nondimensional control with $(I_1, I_2, I_3) = (1, 2, 3)$ and initial $\boldsymbol{\Omega} = (0.03, 1, 0.04)$, integration over 60 time units with fourth-order Runge–Kutta and fine step $h = 0.01$ gives maximum relative drifts below 6.61×10^{-12} for both quantities. The intermediate-axis component changes sign and reaches about -1.00045. Comparing the coarse step $h = 0.02$ with the fine step $h = 0.01$ changes endpoint components by less than 4.02×10^{-10}. This checks the rigid-body baseline. Full attitude reconstruction and any additional acceleration of a falling rotating body have not been computed in this control. This baseline belongs only to the conditional classical comparison. Its success does not detect a medium and is not a prerequisite imposed on the cavity proposal.